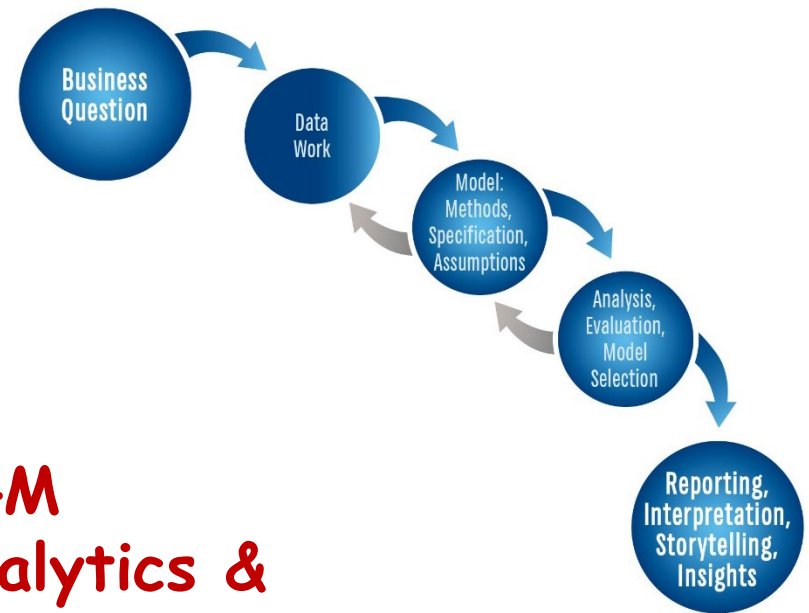




PAML4M

Predictive Analytics & Machine Learning for Managers

3. Basic Models

PAML4M_3R_BasicModels.Qmd

Prof. J. Alberto Espinosa

Last updated 8/1/2026





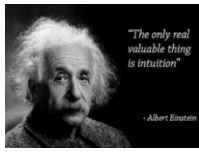
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Predictive Analytics

3.1 OLS Assumptions

See Predictive Modeling Summary on Canvas:
[Predictive Modeling Summary.pdf](#)

OLS Regression Assumptions



OLS Assumptions: Intuition

- How does **OLS** find a regression line that **minimizes the SSE**?
→ The coefficient vector is obtained with this **matrix** operation:

$$\beta = (X' X)^{-1} X' Y$$

- It can be shown **mathematically** that this matrix operation yields the coefficients of the **OLS** regression
- This matrix operation may seem complicated, but it can be computed easily with **matrix** manipulation **software** like **R & Python**
- But this formula is derived from a complex mathematical **proof** that **simplifies** substantially with some simplifying **assumptions**.
- If some of these assumptions are **not met**, the above formula does not necessarily produce the best set of unbiased coefficients for a regression line that minimizes the SSE.
- **Other methods** may be more appropriate in such cases

It's Good to be BLUE

When the **OLS assumptions** are met, the OLS estimators are said to be the:

“Best Linear Unbiased Estimator” (BLUE)

- ✓ **Best** → Estimators have the **lowest variance**
 - ✓ **Linear** → The regression is a linear combination of variables and coefficients.
 - ✓ **Unbiased** → the estimated β coefficients represent the **true effect** (remember that some models yield biased coefficients, for example, if important variables are omitted from the model).
- Need to **test** if **OLS assumptions** are met

Parametric vs. Non-Parametric Methods

- **Parametric** methods: are good for **interpretation**. They are based on **assumptions** made about parameters of the model, which impose mathematical **constraints** to **simplify** the model **estimation**, which is only correct when the parametric assumptions are met.
 - For example: the outcome variable should have a Normal (**OLS**), Poisson (**count data**) or Binomial (**Logistic**) distribution. Y should have a linear relationship with the X's (**OLS**). More restrictive models require the predictors to be normally distributed (**Linear Discriminant Analysis**).
 - **Non-Parametric** methods: are good for **predictive accuracy** when their CV test error is small. They **don't** rely on **assumptions** or parametric constraints and are useful when know little about the data (e.g., **K-Nearest Neighbors**, **Decision Trees**, **Support Vector Machines**, **Neural Networks**).
- **OLS** is a **Parametric** Method → Relies on several **assumptions**

OLS Regression Assumptions

1. (**YC**) The outcome variable (Y) is continuous
2. (**EN**) Errors (residuals) are normally distributed
3. (**XI**) The predictors (X 's) are independent (i.e., uncorrelated)
4. (**LI**) The outcome (Y) and predictor (X 's) variables have a linear relationship
5. (**OI**) (Data) Observations are independent
6. (**EI**) Errors (residuals) are independent
7. (**EA**) The average of the errors (residuals) = 0
8. (**EV**) The variance of the errors (residuals) is constant (errors are "homoscedastic")

OLS Assumption 1

- **YC → Y is continuous** – if Y is **binary**, **categorical**, **discrete** or **truncated**, we need to use other methods (e.g., Logistic, Probit, Linear Discriminant Analysis, etc.)
 - **Test:** observe Y values and check for distributions
 - **Corrections:**
 - ✓ Use the **Generalized Linear Model (GLM)**, rather than **OLS** method with the appropriate distribution family (e.g., binomial, poisson, etc.) – **more on this later**
 - ✓ Use other **classification methods** (e.g., classification trees, linear or quadratic discriminant analysis, etc.)
 - ✓ If Y is **binary** or **categorical** use **Logistic regression**, but ensure that all **other OLS assumptions** hold

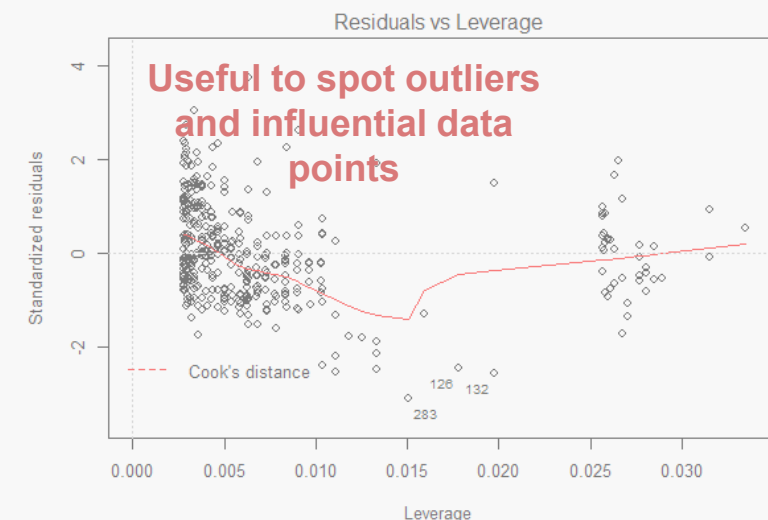
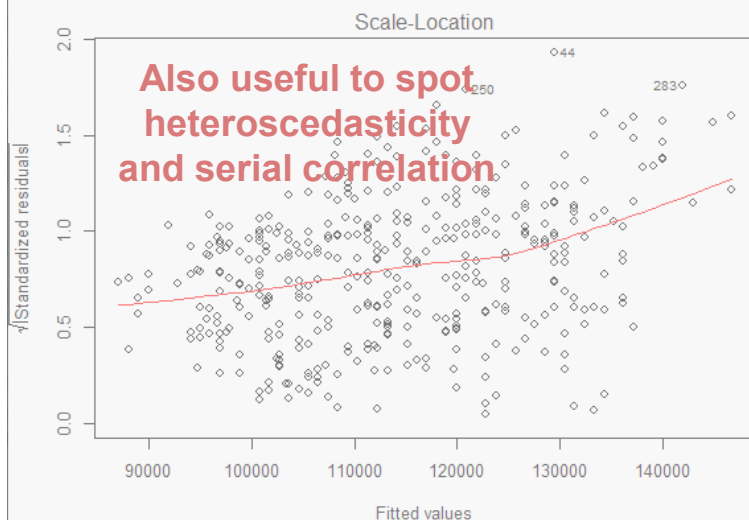
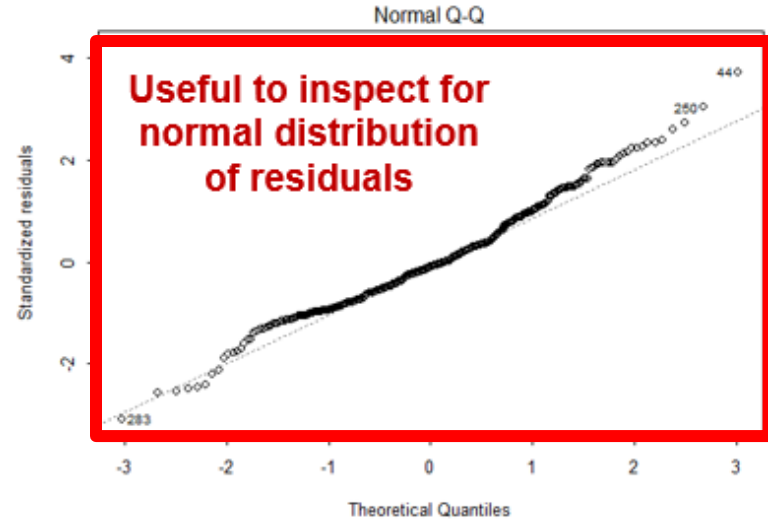
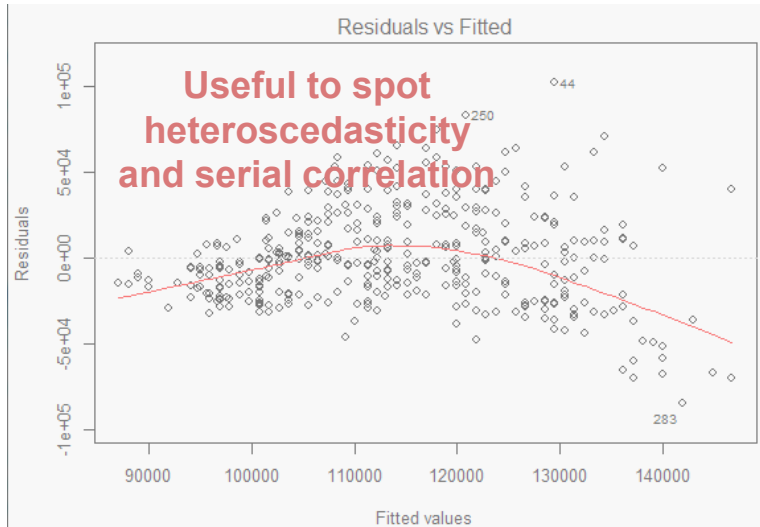
OLS Assumption 2

- **EN** → **Errors (ϵ) are normally distributed** – there are acceptable transformations when errors (i.e., **residuals**) are not normally distributed. Generally (not always), if **Y** is (not) normally distributed → **ϵ** is not either
- **Test:** residual plots; QQ-Plots; normality tests
 - **Correction:** transform Y to achieve normal distribution [e.g., **log(Y)**, Box-Cox, 1/Y, quadratic, etc.]



Regression Plots

QQ-Plot → Visual inspection for normally distributed residuals



OLS Assumption 3

- **XI** → **X's are independent** (uncorrelated) – some correlation in the X's are tolerable, as long as **multi-collinearity** is not **severe**.
 - **Test:** Condition Index (**CI**) and Variance Inflation Factors (**VIF**)
 - **Correction:** With **severe multi-collinearity**, other methods are more appropriate (e.g., Ridge, LASSO, Principal Components Regression PCR, Partial Least Squares PLS, etc.)

OLS Assumption 4

- **LI** → **Y and X's have linear relationship** – if not, some X's can be transformed (log, squared, etc.) to create a linear model
 - **Test:** scatter plots; compare linear to non-linear model fit statistics
 - **Correction:** use non-linear models (e.g., interactions, polynomials, piecewise models, splines, etc.)

OLS Assumptions 5 & 6

- **OI → Observations are independent** – if one observation is influenced by another (e.g., a stock price today will be correlated with its price tomorrow; survey responses by members of a family or association).
- **EI → Errors are independent** – if errors are correlated it is an indication that there is a missing variable in the model or there is serial correlation. If **OI** does not hold, usually neither does **EI**
 - **Test:** serial or auto correlation (e.g., **Durbin-Watson**)
 - **Corrections:**
 - ✓ Use **forecasting**, time series, causal models, or other methods that correct for **serial correlation**
 - ✓ **Lagging** the outcome variable or the predictors often helps reduce serial or auto correlation.

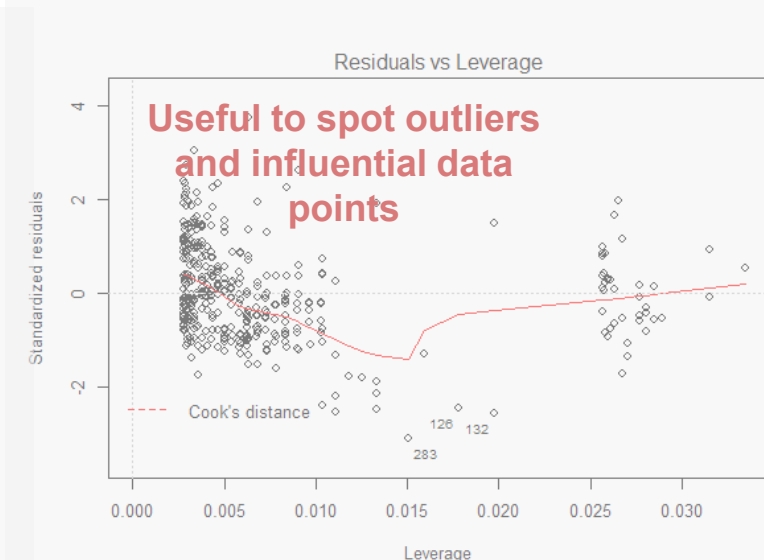
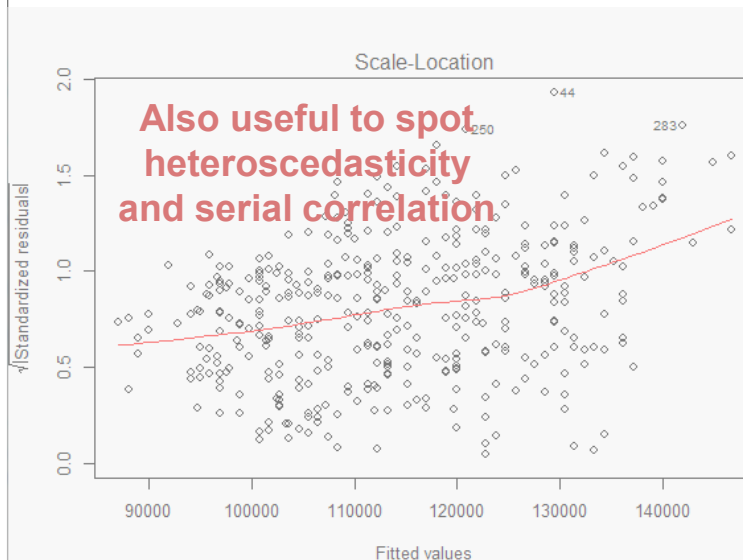
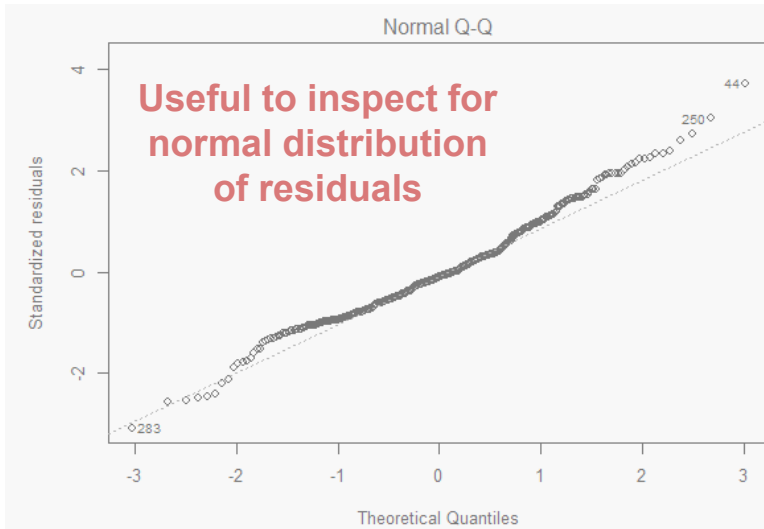
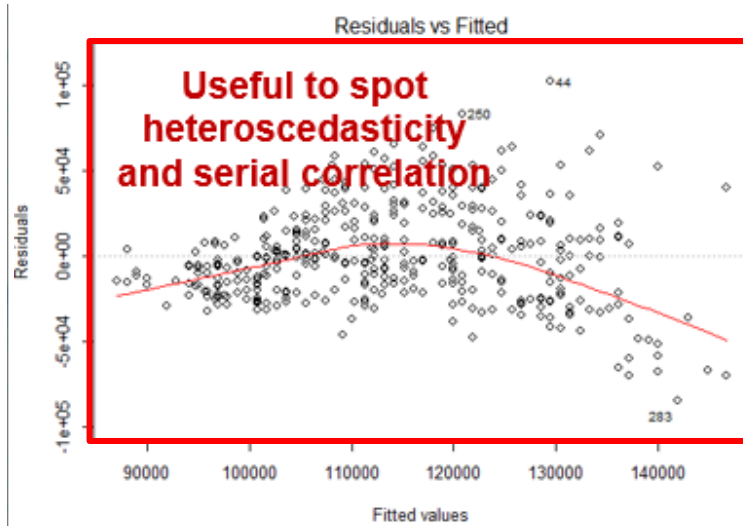
OLS Assumptions 7 & 8

- **EA → The error average is 0** (+/– errors average out)
 - **Test:** sum or average the residuals → not necessary if the model has an intercept → OLS calculations take care of this
- **EV → The error variance is constant (homoscedastic)** – uneven residuals (**heteroscedastic**) cause observations with large errors to pull the regression strongly in both directions, thus increasing variance.
 - **Test:** inspect residual plots, conduct heteroscedasticity tests (e.g., **Breusch-Pagan Test**)
 - **Correction:** use robust methods, **Weighted Least Squares (WLS)**, etc.



Regression Plots

Residual Plot → Visual inspection for Heteroskedasticity and Serial Correlation



Predictor Assumptions in OLS

- With a **large samples** of **50 degrees of freedom** (number of observations, minus number of predictors, minus 1), the **predictors don't** have to be normally distributed to use **OLS**
- Predictors can be **counts, binary, categorical**, etc.
- With **small samples**, the **predictors** must be **normally distributed**
- In contrast, other modeling methods like Linear Discriminant Analysis (**LDA**), Quadratic Discriminant Analysis (**QDA**), do indeed require **predictors** to be **normally distributed**.



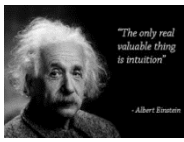
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Predictive Analytics

3.2 Weighted Least Squares (WLS)

Heteroskedasticity:

EV: error (residual) variance is not constant



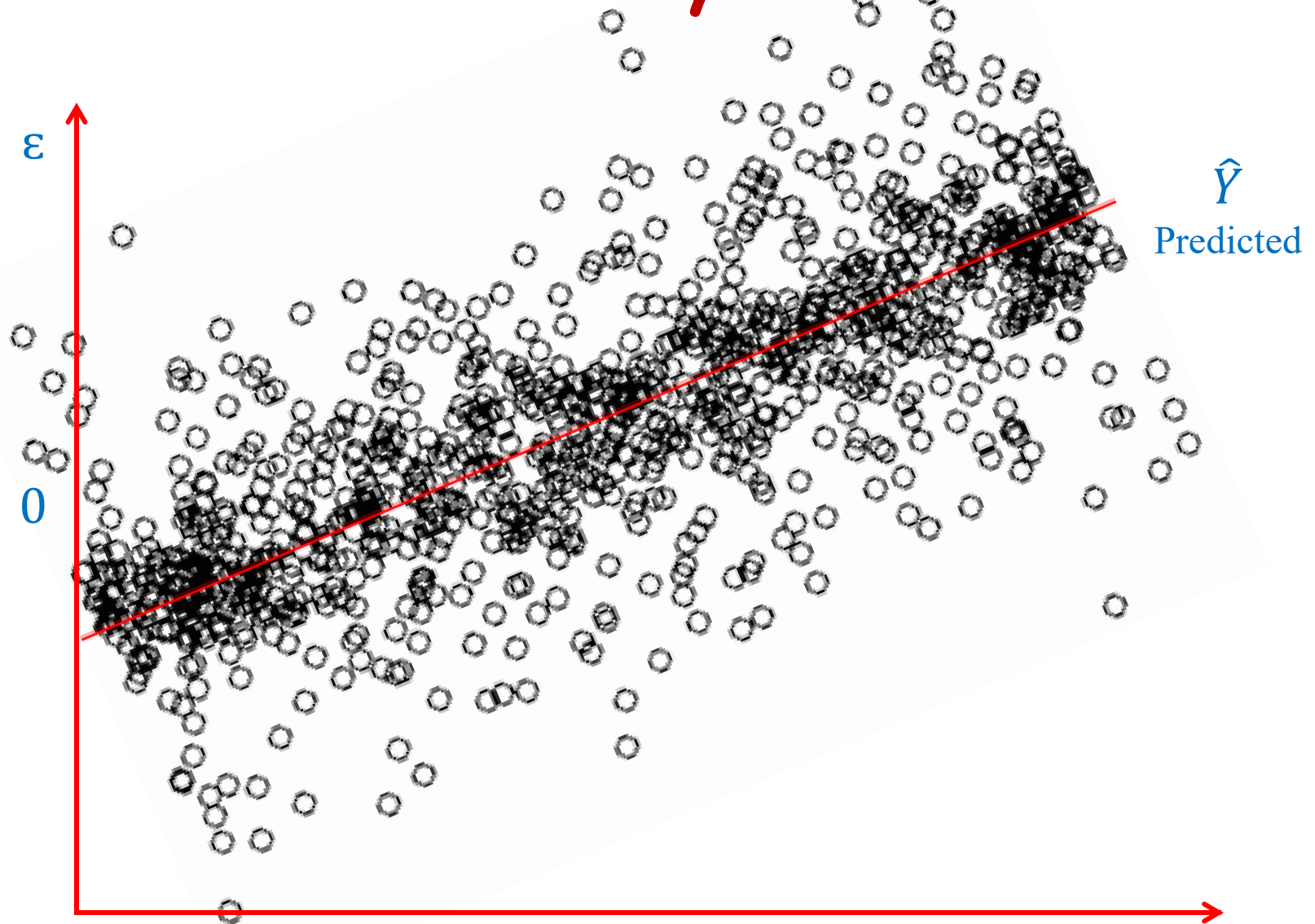
Heteroscedasticity: Intuition

- The **OLS** estimation method fits a line that **minimizes the SSE**.
- But if the errors **grow** or **shrink** systematically along the outcome variable it causes the squared errors to be too large (or too small) in some parts of the regression line.
- If so, **OLS** is still **unbiased**
- But **OLS** is no longer **most efficient** (least variance)
- This problem of “**uneven**” error variance is referred to as “**heteroscedasticity**” or “**non-spherical residuals**”

Detecting Heteroscedasticity

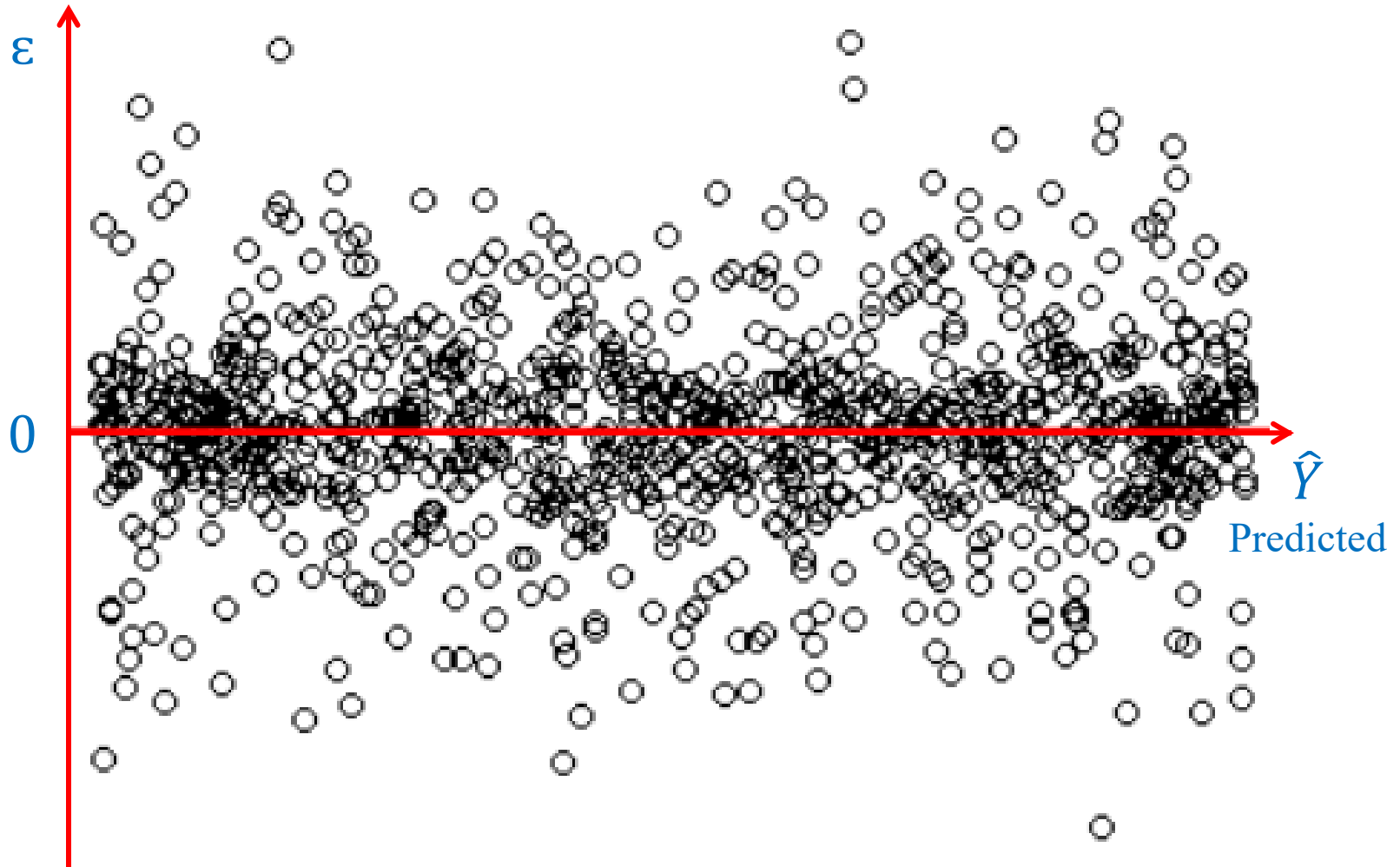
- You can do a visual inspection of the **residual plot** ($\hat{Y}$ vs. ε) $\rightarrow$ the errors should look like an even cloud (i.e., “**homoscedastic**”)
- If not $\rightarrow$ the errors are “**heteroscedastic**”
- If you have a **business reason** to believe that errors increase or decrease with one or more variables (e.g., as people learn they make smaller errors; the closer to the city the more unpredictable the traffic) you should inspect for heteroscedasticity.
- If you suspect heteroscedasticity in the errors, you should **test** it

Homoscedasticity Illustration

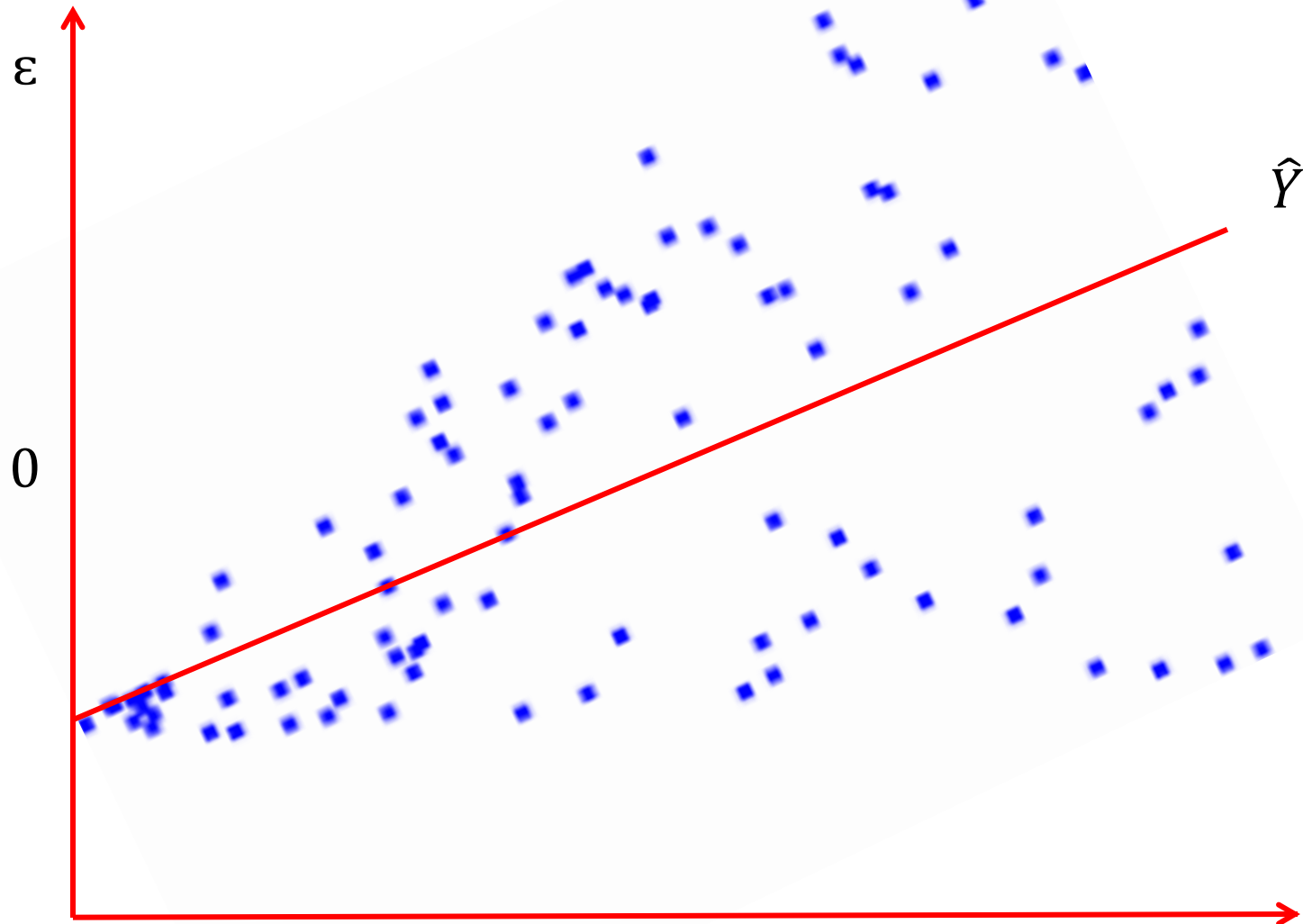


Homoscedasticity Illustration

Residual Plot (vs. Predicted Values)

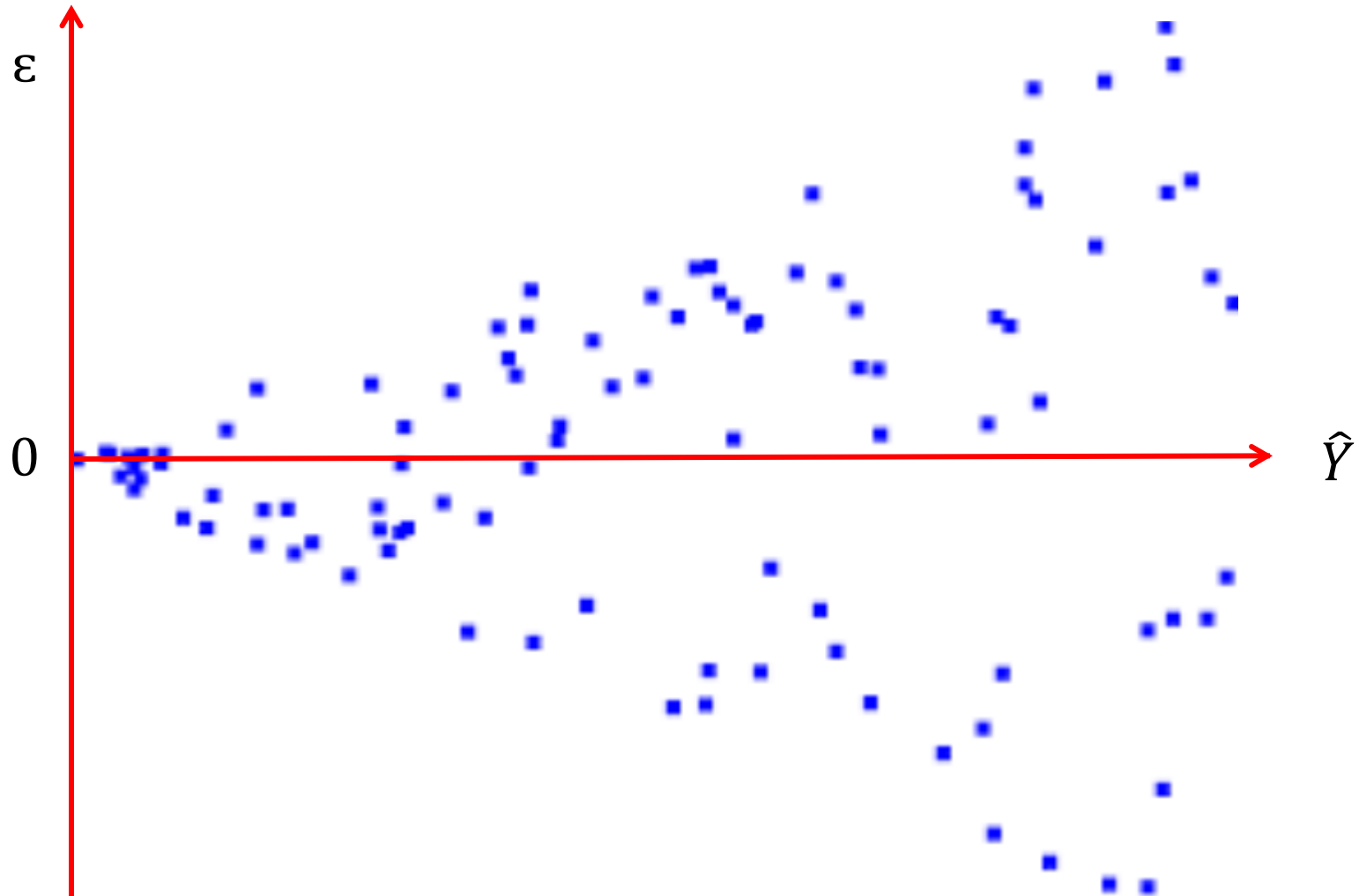


Heteroscedasticity Illustration



Heteroscedasticity Illustration

Residual Plot (vs. Predicted Values)



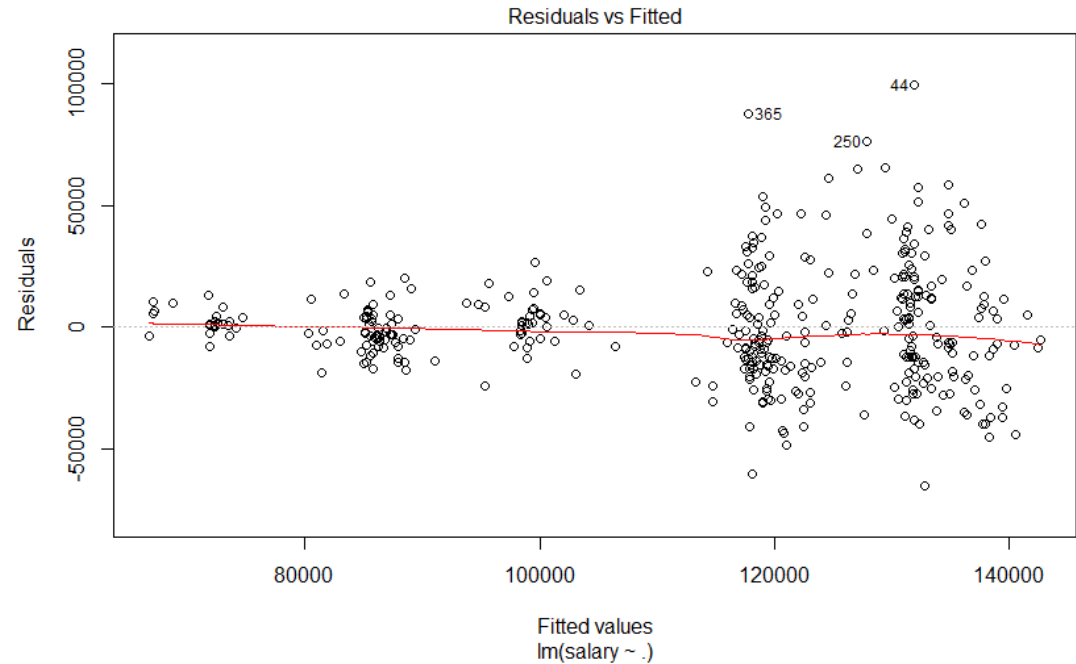
Testing for Heteroscedasticity

- Heteroscedasticity is relatively **easy to identify**
- After fitting an OLS model, inspect the residual **plot** against predicted values. The **residuals** should look like an even **cloud**
- If not, then **test** for **heteroskedasticity**
- There are various heteroskedasticity tests → the most popular ones use **regression** methods, e.g.,
 - Regress the squared **residuals** on the **predicted values**
 - $\varepsilon^2 = \beta_0 + \beta_{\hat{Y}}(\hat{Y})$
 - **Breusch-Pagan Test:** if the **p-value** of the residual regression is **significant**, then the errors are correlated with the predicted values → **heteroscedasticity** is present (i.e., Reject H_0 of **no heteroscedasticity**)
 - **White's Test:** examine the **R^2** of the residual regression
- Naturally, **R** does all of this for us very easily

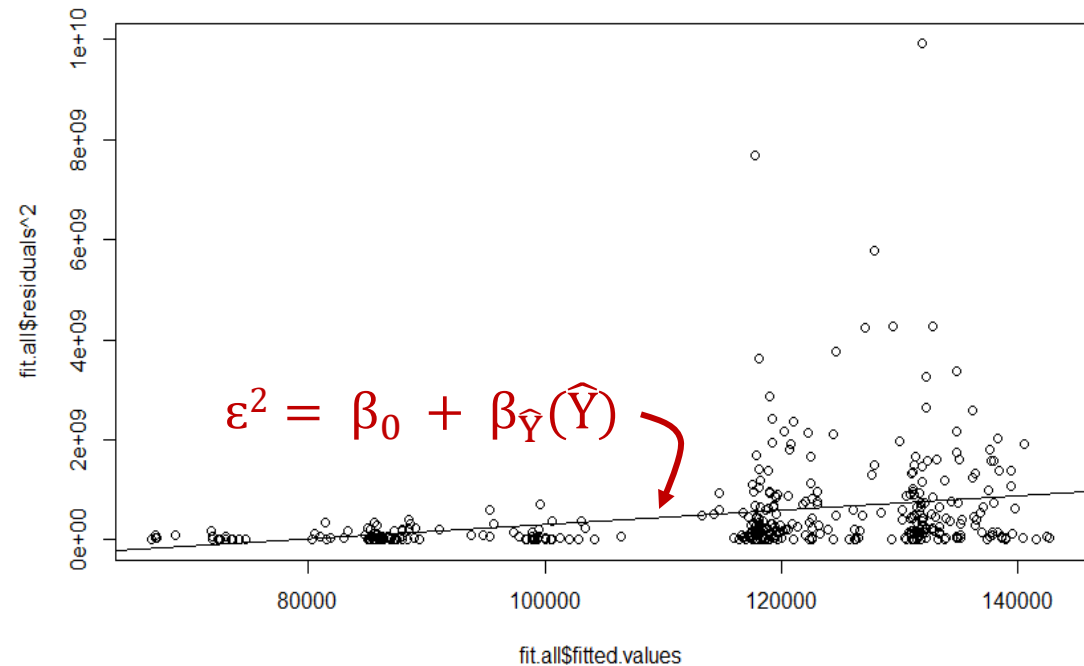
Heteroscedasticity Illustration

`fit.all <- lm(salary~., data = Salaries)`
see Homework 2

Plot (Predicted vs. Residuals) →



Plot (Predicted vs. Residuals²) →



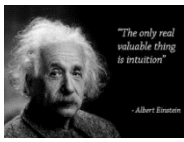


Breusch-Pagan Test for Heteroskedasticity

- {library} Topic
Search keywords
- {lmtest} BP Test for Heteroskedasticity
`bptest(lm.ols, data = Hitters)`

Weighted Least Squares (WLS)

Correcting for heteroskedasticity



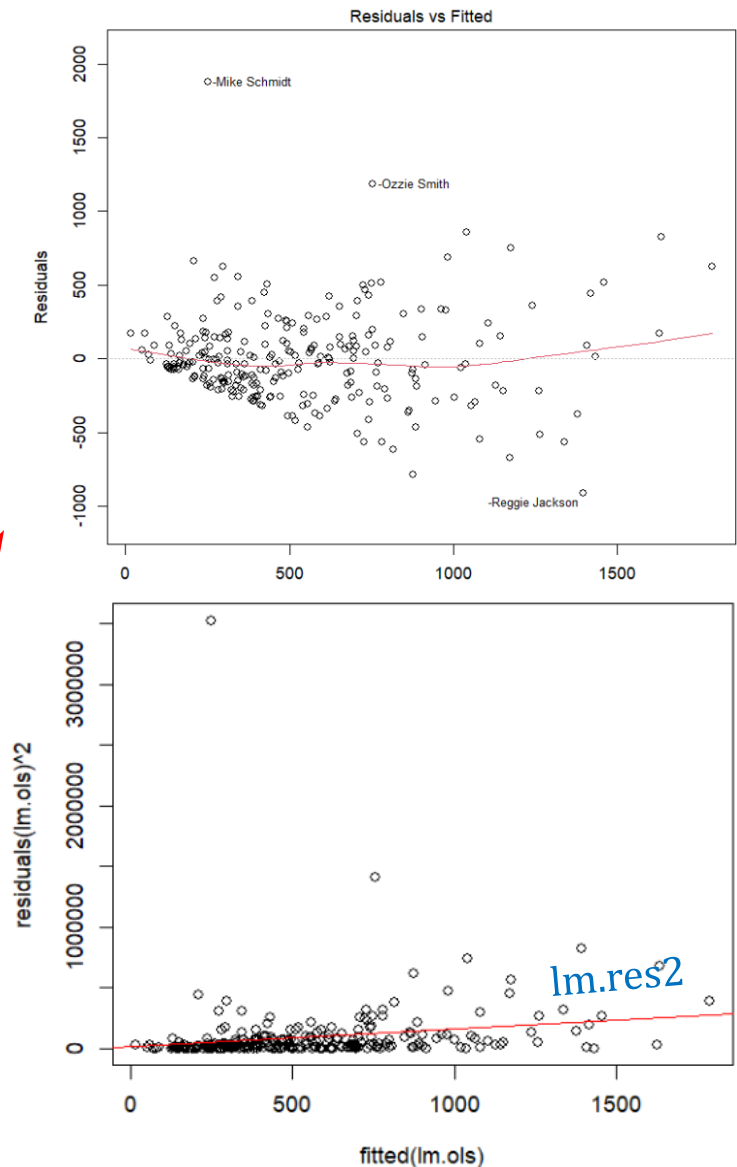
Weighted Least Squares (WLS): Intuition

- Regression models with **heteroskedastic** residuals can be easily **corrected** with **WLS**
- **WLS** is a generic regression estimation method that minimizes a **weighted sum of error squares**, rather than just the **sum of error squares**
- The idea is to give **less weight** to data points with large residuals, and more weight to those with small residuals
- If OLS assumptions (other than EV) hold, both **OLS** and **WLS** are **unbiased** → i.e., their coefficients are correct on average.
- But with heteroskedasticity **WLS** has less **variance** → the regression is more **stable** with more **significant** coefficients
- Many **weighting** methods are available for **WLS**, but the most popular and effective approach is to use the **OLS residuals**

Fitting a WLS Model

A multi-step process:

1. Fit a plain **OLS** regression and extract the residuals ϵ and predicted values $\hat{y}$ 
2. Fit another model to regress the **squared values** of the **OLS residuals** on the **OLS predicted values** $\rightarrow \text{lm.res2} <- \epsilon^2 \sim \hat{y}$ 
3. Then use the **inverse squared** of the resulting **fitted** (predicted) values of this ϵ^2 model to create a weight vector $\rightarrow \text{wts} = 1 / \text{fitted}(\text{lm.res2})$
4. Fit a (weighted OLS =) **WLS** regression with parameter **weights = wts**





WLS Regression

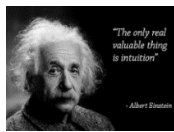
- {library} Topic
Search keywords
- {lmtest} BP Test for Heteroskedasticity
`lm.res2 <- lm(residuals(lm.ols) ^ 2 ~ fitted(lm.ols))`

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Predictive Analytics

3.3 Generalized Linear Models (GLM)

Maximum Likelihood Estimation (MLE)



Maximum Likelihood: Intuition

- **Maximum Likelihood Estimation (MLE)** seeks to maximize the **likelihood** that a model fits the data accurately
- That is, it seeks to fit a model that has a good level of “**agreement**” between what the model **predicts** and the **actual** outcomes in the data
- For each observation, the **likelihood function** provides the likelihood (i.e., probability) that the respective **outcome** is **predicted** correctly – i.e., the correct **value** or correct **classification**

Maximum Likelihood: Details

- A regression estimated with **MLE** (rather than OLS) finds the set of β coefficients that **maximizes** the **likelihood** of a correct prediction
- As such **MLE** provides more **precise estimates** when the outcome variable is **not normally distributed** (e.g., **Poisson**, **Logit**) or the **outcome** is **not continuous** (e.g., **binary**, **categorical**, **counts**)
- However, if the outcome variable is continuous and normally distributed, **MLE** and **OLS** yield the exact **same** results

Generalized Linear Models (GLM)

OLS **EN**(x)

Use when errors or Y are not normally distributed

Generalized Linear Models (GLM)

- OLS models are fit by finding the line that minimizes SSE
 - R fits OLS regressions with the “Linear Model” `lm()` function, which should only be used when OLS assumptions are met.
-
- But if the outcome variable or the residuals don’t follow a normal distribution, GLM is a better modeling method.
 - In R, GLM is fitted with the `glm()` function
 - The `glm()` syntax is similar to `lm()`, but you also need to specify the type of distribution of the outcome variable and a link function.

GLM Regression Assumptions

1. ~~(VC) The outcome variable (Y) is continuous~~
Not a GLM assumption
Modified from OLS
Same as with OLS
2. (ED) Errors (residuals) and Y are NOT required to be normally distributed, but need to follow a **Distribution Family** (e.g., Gaussian, Poison, Binomial, Multinomial)
3. (XI) The predictors (X's) are independent (i.e., uncorrelated)
4. (LL) The **transformed outcome** variable (not its raw value) the outcome by the **link** function and the **predictor** variables have a linear relationship (e.g. $\text{logit}(Y) \sim X$)
5. (OI) (Data) Observations are independent
6. (EI) Errors (residuals) are independent
7. (EA) The average of the errors (residuals) = 0
8. ~~(EV) The variance of the errors (residuals) is constant
—— (errors are "homoscedastic")~~

Fitting GLM Models

- **How GLM fits models:**

- GLM fits many regression **lines** through complex **algorithms**
- GLM then calculates a **likelihood function** for each **line**
- GLM then **selects** the line that **maximizes** the **likelihood function** → that is, the line that **minimizes**:

$$\text{Deviance or } 2LL = (-2 * \text{Log Likelihood})$$

- **Likelihood Function** → multiply the likelihood of a correct prediction for each data point
 - **MLE** finds the linear model in which this **likelihood** is **maximized**



-2 Log Likelihood (2LL) Explained

- Why the Log ?

It is a mathematical **convenience**. A **likelihood** is bound between 0 and 1 → so its **log** will be between $-\infty$ and 0

- Why the – ?

- ✓ Likelihoods are 0 to 1 → so log-likelihoods are **negative**
- ✓ So, the – makes the log-likelihood values **positive**
- ✓ **Maximizing** a **likelihood** value is the same as **minimizing** its **negative** value → so its **interpretation** is similar to **SSE** → the **model** with **smallest 2LL (Deviance)** is the **best** !!

- Why the 2 ?

Another mathematical **convenience** →

- ✓ $\text{Log}(X^2) = 2 * \text{Log}(X)$
- ✓ **2 * Log Likelihood** has a X^2 (Chi-Square) **distribution**, which is convenient for model comparison and testing.

2LL = Deviance

- Just like ANOVA or MSE for quantitative models, 2LL can be used to:
 1. Evaluate a Single Model → Smaller Deviance is better; or
 2. Compare 2 nested models → Log-Likelihood ratio test $2LL_R$:

$$2LL_R = -2 * \text{Log}\left(\frac{L_{\text{Small}}}{L_{\text{Large}}}\right) = (-2) * \text{Log}(L_{\text{Small}}) + (-2) * \text{Log}(L_{\text{Large}})$$
$$= \text{Deviance}_{\text{small}} - \text{Deviance}_{\text{Large}}$$

- ✓ A small $2LL_R$ means that the 2 models are similar, so the Reduced Model is preferred (i.e., simpler)
 - ✓ A large $2LL_R$ means that the Large Model is preferred → it reduces the deviance substantially
- OLS and MLE select the same model when OLS assumptions hold



Example: GLM Regression

- {library} Topic
Search keywords
- {MASS} Fitting an OLS model with GLM
`glm.fit = glm(medv ~ lstat + age,`

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Predictive Analytics

3.4 Logistic Regression

Odds and Probabilities

Odds and Probabilities

- $P_{Y=1}$ = Probability that $Y = 1$ (success; e.g., approving a loan)
- $P_{Y=0} = 1 - P_{Y=1}$ = Probability that $Y = 0$ (failure; e.g., declining a loan)
- Odds is defined as the ratio of the probability of success to the probability of failure.
- Odds of success (or $Y=1$) = $\frac{P_{Y=1}}{P_{Y=0}} = \frac{P_{Y=1}}{1-P_{Y=1}}$
- For example, if the probability of winning a baseball game is 60% or 0.60, the odds of winning is 0.60/0.40, or 3 to 2, or 1.5
- $P < 0.50 \rightarrow O < 1$; $P = 0.50 \rightarrow O = 1$; $P > 0.5 \rightarrow O > 1$

Log-Odds: The Logit Function

- The **Logit** function is an important transformation in predictive modeling for **classification models**
- **Logit** functions can be **binomial** (i.e., 2 possible values, e.g., 0 or 1, yes or no, success or failure, etc.) or **multinomial** (> 2 categories (e.g., small, medium, large))
- More on **multinomial** logit later
- The **Logit function** is a special type of **transformation** (for statistical convenience) in which the outcome variable is modeled as the **Log(Odds)** that **$Y = 1$ (success)**
- **$\text{Logit}(Y) = \text{Log-Odds}$** of success (of **$Y = 1$**) =

$$\text{Log}\left(\frac{P_{Y=1}}{P_{Y=0}}\right) = \text{Log}\left(\frac{P_{Y=1}}{1 - P_{Y=1}}\right)$$

Conversion Examples

- If the **probability** that the Yankees will **win** a game P_W is **0.25** or **25%**

- Then the **probability** of **losing** is

$$P_L = 1 - 0.25 = 0.75 = 75\%$$

- The **odds** of winning are

$$\frac{P_W}{P_L} = \frac{0.25}{0.75} = 0.33 = \frac{1}{3} \text{ or } 1 \text{ to } 3$$

- The **log-odds** of winning are: $\text{Log}(0.33) = -1.109$

- The **odds** of winning are $e^{-1.109} = 0.33$

Logistic (Logit) Models

OLS $YC(x)$

Use when Y is categorical (not continuous) →

- (1) Only 2 categories → Binomial Logit
- (2) More than 2 categories → Multinomial Logit

The Logistic Regression Model

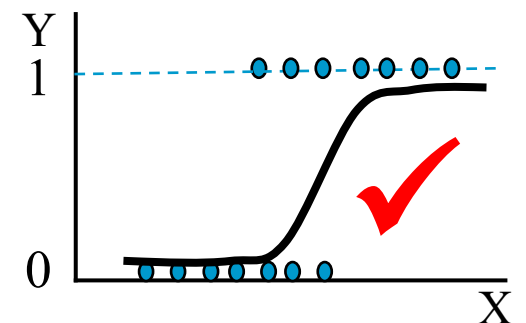
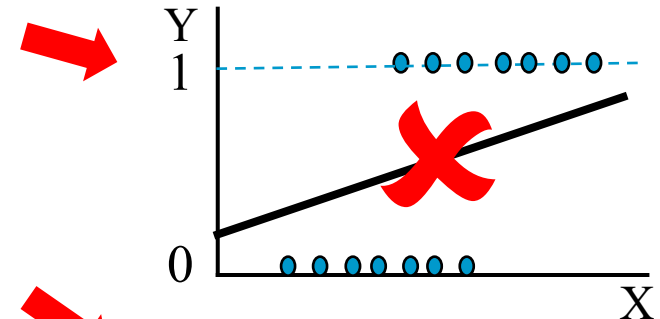
- The dependent variable Y can only be 0 or 1 (not continuous $\rightarrow$ must transform)

- Transform Y into $\text{Logit}(Y) = \text{Log}\left(\frac{P_{Y=1}}{1-P_{Y=1}}\right)$

The X 's are **not** transformed:

$$\text{Logit}(Y) = \beta_0 + \beta_1(X_1) + \beta_2(X_2) + \beta_3(X_3) + \dots + \varepsilon$$

- Fit the model with the $\text{glm}()$ (GLM), rather than the $\text{lm}()$ (OLS) model
- The $\text{lm}()$ model will yield similar results if you **manually transform** Y into $\text{Logit}(Y)$, but the $\text{glm}()$ is the correct method and its fit statistics are the most appropriate.



Logistic Model Interpretation

- The **OLS** regression coefficients β show how much Y changes when the corresponding X increases by 1

- In contrast, the **Logistic** regression coefficients β show, on average and holding everything else constant, how much the **Log-Odds** of $Y = 1$ (e.g., success) **change** when the corresponding X increases by 1

- But the **Log-Odds** effects are difficult explain. What is the managerial meaning of **Log-Odds** effects?
- For managerial interpretation, it is best to convert **Log-Odds** effects into **Odds** effects $\rightarrow \text{Odds} = e^{\text{Log-Odds}}$

- The interpretation of odds effects $\text{Odds}(\beta) = e^{\beta}$ is multiplicative: when X increases by 1 the odds coefficient e^{β} indicates that Y changes by a **factor** of $\beta \rightarrow$
 - log-odds **(-)** $\rightarrow$ odds $< 1 \rightarrow$ **reduce** the odds of Y being 1
 - log-odds $= 0 \rightarrow$ odds 1 $\rightarrow$ **no change** in odds
 - log-odds **(+)** $\rightarrow$ odds > 1 **increase** the odds of Y being 1

Logistic Regression Assumptions

1. (YC) The outcome variable (Y) is categorical
(**binary** or **multinomial**, not continuous) ~~Not a Logit assumption~~
Modified from OLS
Same as with OLS
2. ~~(EN) Errors (residuals) are normally distributed~~
3. (XI) The predictors (X's) are independent (i.e., uncorrelated)
4. (LL) The outcome (**Logit of Y**) and predictor (X's) variables have a linear relationship
5. (OI) (Data) Observations are independent
6. ~~(EI) Errors (residuals) are independent~~
7. ~~(EA) The average of the errors (residuals) = 0~~
8. ~~(EV) The variance of the errors (residuals) is constant
—— (errors are "homoscedastic")~~

Fun Facts about Log-Odds

- If $P_{Y=1} > 0.5$ (50%) $\rightarrow$ Odds $_{Y=1} > 1 \rightarrow$ Log-Odds $_{Y=1} > 0$ (positive)
- If $P_{Y=1} < 0.5$ (50%) $\rightarrow$ Odds $_{Y=1} < 1 \rightarrow$ Log-Odds $_{Y=1} < 0$ (negative)
- The fact that the Odds increase when the Log-Odds are positive and that they decrease when the Log-Odds are negative, is a very convenient feature of Log-Odds for interpretation.
- It is also very easy to **convert** probabilities to/from odds and log-odds
 - ✓ Odds = $\frac{\text{Prob}}{1 - \text{Prob}} \rightarrow \text{Log-Odds} = \text{Log}(\text{Odds})$
 - ✓ Prob = $\frac{\text{Odds}}{1 + \text{Odds}} \rightarrow \text{Odds} = e^{\text{Log}(\text{Odds})} = \exp(\text{Log-Odds})$

More Fun Facts About Logs

[Aside] Fun Facts about Logs

- Logarithms have a base $\rightarrow$ If $b^x = y \rightarrow \text{Log}_b(y) = x$
- That is, the Log of y on base b is x
- So, if $y = 10^x$, then $\log_{10}(y) = x \rightarrow$ the Log of y on base 10 is x
- If $y = e^x$, then $\log_e(y) = \text{LN}(y) = x$
 - That is, the Natural Log of y (on base e) is x
 - Where e is the Euler number = 2.718 ..., widely used in math
- All logs in statistics are natural logs (on base e)
- So, when we write $\log(x)$ it is implied that it is base e or LN

What is e any way?

- It is your professor's last name, but also **Euler's**
- Interestingly, it can be computed in many ways
- For example, \$1 invested at an annual interest rate of 100% yields $(1+1) = \$2$ in one year
- If interest is compounded monthly it yields $(1+1/12)^{12} = \$2.61$
- Compounded daily yields $(1+1/365)^{365} = \$2.714$
- As the number of compounding periods gets larger (∞), the yield approximates $e = (1+1/\infty)^\infty = 2.718 \dots$
- Interestingly e is also $= 1/0! + 1/1! + 1/2! + 1/3! + \text{etc.}$

Useful Facts about Logs

- $\text{Log}(a * b) = \text{Log}(a) + \text{Log}(b)$
- $\text{Log}(a / b) = \text{Log}(a) - \text{Log}(b)$
- $\text{Log}(x^2) = 2 * \text{Log}(x)$ and $\text{Log}(x^N) = N * \text{Log}(x)$
- Negatives have no logs
- $\text{Log}(0 < x < 1) = (-)$
- $\text{Log}(1) = 0$
- $\text{Log}(x > 1) = (+)$
- $e^{\text{Log}_e(x)} = x$
- So, $e^{\text{LogOdds}} = \text{Odds}$ and $\text{Log}_e(\text{Odds}) = \text{LogOdds}$
- In  $\rightarrow \exp(\text{LogOdds}) = \text{Odds}$
- $\log(\text{Odds}) = \text{LogOdds}$

Log Conversions

- Log Odds (0) $\rightarrow$ Odds = 1 $\rightarrow$ Prob = 0.5
 $\rightarrow$ chance (flip a coin)
- If $\beta = 0$ (i.e., not significant) the predictor doesn't change the odds
- If Log Odds (-) $\rightarrow$ Odds < 1 $\rightarrow$ Prob < 0.5
 $\rightarrow$ (-) β in a logistic regression
 $\rightarrow$ The log odds go down
 $\rightarrow$ The odds go down by a (<1) factor
- If Log Odds (+) $\rightarrow$ Odds > 1 $\rightarrow$ Prob > 0.5
 $\rightarrow$ (+) β in a logistic regression
 $\rightarrow$ The log odds go up
 $\rightarrow$ The odds go up by a (>1) factor

Prob	Prob%	Odds	Log Odds
		$p / (1-p)$	$\ln(o)$
0.001	0.1%	0.001	-6.907
0.01	1%	0.010	-4.595
0.1	10%	0.111	-2.197
0.2	20%	0.250	-1.386
0.3	30%	0.429	-0.847
0.4	40%	0.667	-0.405
0.5	50%	1.000	0.000
0.6	60%	1.500	0.405
0.7	70%	2.333	0.847
0.8	80%	4.000	1.386
0.9	90%	9.000	2.197
1.0	100%		

Log Odds	Odds	Prob	Prob%
	$e^{(\text{Log Odds})}$	$o / (1+o)$	
-6.907	0.001	0.001	0%
-4.595	0.010	0.01	1%
-2.197	0.111	0.1	10%
-1.386	0.250	0.2	20%
-0.847	0.429	0.3	30%
-0.405	0.667	0.4	40%
0.000	1.000	0.5	50%
0.405	1.500	0.6	60%
0.847	2.333	0.7	70%
1.386	4.000	0.8	80%
2.197	9.000	0.9	90%
		1.0	100%



Logistic Regression

- `{library}` Topic
Search keywords
- Binomial Logistic Regression
`heart <- read.table("Heart.csv",`
`null.dev <- paste("Null Deviance",`
- Log-Odds and Odds Coefficients
`cbind("Log Odds" = lm.sum$coefficients[ ,1],`

ITEC 621

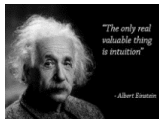
Predictive Analytics

3.5 Decision Trees

Analytics Modeling Options

	Modeling Method		
	Structured		Visual, Text, Unstructured, etc.
Descriptive	Cluster analysis, correlation, market basket analysis, sample statistics, ANOVA		Bubble charts, network diagrams, natural language processing, clustering dendograms, etc.
Predictive	Association	Decision Tree	Charts
Quantitative Value	Regression	Regression Trees	Regression plots, scatter plots, Tableau diagrams, trend charts, etc.
Classification	Logistic Regression; Other Categorical Regression Models	Classification Trees	Tree maps, interactive diagrams,
Prescriptive	Operations research, decision modeling, optimization, linear programming		Simulations, etc.

Decision Trees Overview



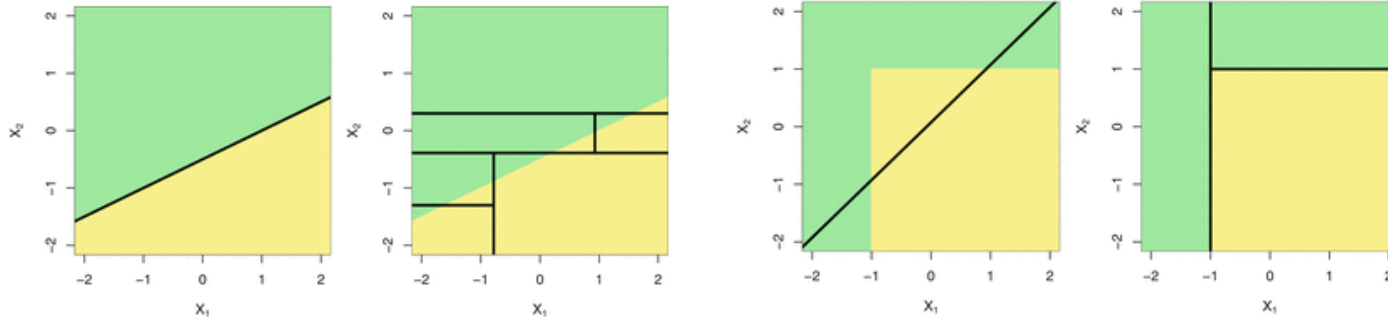
Decision Trees: Intuition

- Trees are **non-parametric** models → no need to check assumptions
- There are **two types** of decision trees:
 - **Regression Trees** – the outcome is **quantitative** → solve similar problems than linear **regression** models → but **less interpretable**
 - **Classification Trees** – the outcome is **categorical** → solve similar problems than **Logistic**, **LDA** and **QDA** models → **but less interpretable**

Comments on Trees

Both, regression and classification trees are similar in that:

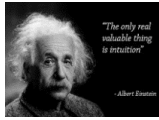
- **Observations** are **split** recursively into nested “**regions**” (**leaves**) to guide the **prediction** of new observations into one of the resulting regions.
- Which model type is **better**, tree or regression? It **depends** !
- The 2 left diagrams below show both models when the “**true**” relationship between 2 variables is **linear**. The 2 **right** diagrams they are **not**.



Regression Trees

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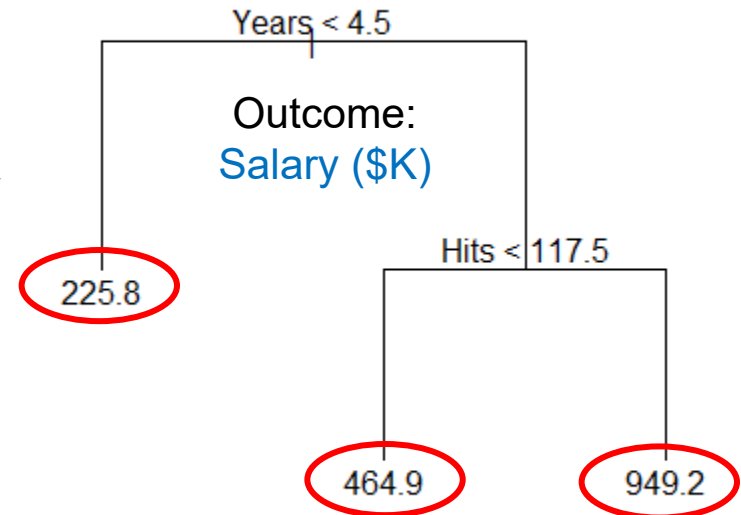


Regression Trees: Intuition

- The intuition is simple. If we have a **quantitative** outcome Y , we find the value of predictor X_1 that **separates** the data into **two groups** whose **outcome means** are the **farthest** in the data and **split**.
- The corresponding value X_1 where the split occurs is called a “**tree node**”, which separates the data into **2** tree sections called “**branches**”.
- We then find the predictor X_2 (or X_1 again) that can separate the outcomes the farthest within each branch; and **so on**.

Regression Tree Illustration

- In the tree **illustration** (see textbook) we are predicting baseball **player Salary** based on the player's number of **years** in the major leagues and number of **hits** per year.
- So, on average, a player with less than **4.5 years** will make a **Salary** of **\$225.8K**
- A player with more than **4.5 years** and more than **117.5 hits** will make a **Salary** of **\$949.2K**
- A player with more than **4.5 years** but with less than **117.5 hits** will make a **Salary** of **\$464.9K**

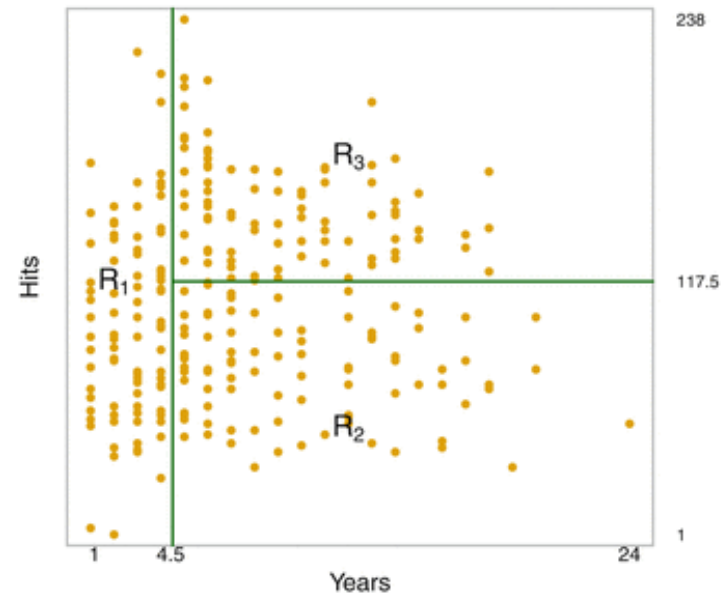


Regression Trees: Explained

- Essentially, we **partition** the data at various **nodes** into “**regions**” such that the distance between data points **within** regions is **minimal** and the distance **between** regions is **largest**.
- Any region can be **further subdivided** into more regions
- All observations within a **region** are assigned the same prediction equal to the **mean** of all outcome values in the region

Tree Regions

- The diagram shows two partitions in the baseball salary example, with the first partition at $\text{Years} < 4.5$ and the region for $\text{Years} \geq 4.5$ further sub-partitioned at $\text{Hits} < 117.5$
- With P predictors, the regression tree method finds the specific predictor x_i and **cutoff values** of that predictor that minimizes the **ESS**
- It then finds which of the **resulting partitions** to split, one at a time, and **where**, such that **ESS** is further reduced and minimized again
- We then continue splitting regions into further sub-regions until **each branch** has exactly **one data point**, although we typically stop growing the tree at some point (more on this later).





Regression Trees

- {library} Topic
Search keywords
- Regression Trees
`tree.boston <- tree(medv ~ ., Boston)`



Growing Trees

When do we Stop Growing a Tree?

- A tree can be grown until every leaf in the tree is **single data point**.
- However, growing large trees is **not very useful** and often you can't even display the resulting tree.
- Later in the course, we will discuss how to **grow** and **prune** trees based on **cross-validation testing**, which will yield the particular tree size that maximizes **predictive accuracy**.
- For now, you can control the size of the tree by changing the **mindev** attribute in the **tree()** function.
- The **default** is **mindev = 0.01** → when the deviance (**MSE**) in a leaf is less than 0.01 or **1%** of the deviance in the **root leaf**, **tree()** stops growing the tree → a further split is no longer worth the deviance reduction.
- **Increase** the value of **mindev** to get **fewer** leaves, **decrease** it to get **more** leaves.



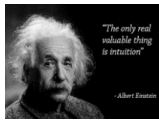
Controlling the Tree Size

- {library} Topic
Search keywords
- Small Tree Example, $\text{mindev} > 0.01$
`tree.boston.small <- tree(medv ~ ., Boston,`
- Large Tree Example, $\text{mindev} < 0.01$
`tree.boston.large <- tree(medv ~ ., Boston,`

Classification Trees

Analytics Modeling Options

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Classification Trees: Intuition

- Classification trees are **similar** in concept to **regression trees**
- Except that the **outcome** is the prediction of a **categorical** outcome from **K** possible **classes**
- **K = 2** → **Binomial** classification trees
- **K > 2** → **Multinomial** classification trees
- The classification is based on the **highest probability** of classifying observation in the **correct class**
- So, it is an **alternative** method to **Logistic Regression**
- In contrast to regression trees, a branch cannot be further subdivided when it contains outcomes of the same category, which is known as a **“terminal node”**.



Classification Trees

- {library} Topic
Search keywords
- Classification Trees (Binomial)
`tree.boston.small <- tree(medv ~ ., Boston,`
- Large Tree Example, `mindev < 0.01`
`tree.boston.large <- tree(medv ~ ., Boston,`